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Ex: Expand $f(z) = \frac{1}{z(z-1)}$ as Laurent's Series in the domain $1 < |z-2| < 2$.

→ We have,

$$f(z) = \frac{1}{z(z-1)}$$

$$= \frac{A}{z} + \frac{B}{z-1} \quad (\text{assume})$$

$$= \frac{A(z-1) + Bz}{z(z-1)}$$

$$\therefore 1 = A(z-1) + Bz$$

Putting $z=1$

$$\therefore 1 = 0 + B$$

$$\boxed{B=1}$$

Putting $z=0$

$$\therefore 1 = A(0-1) + 0$$

$$\therefore 1 = -A$$

$$\boxed{A=-1}$$

$$\therefore f(z) = -\frac{1}{z} + \frac{1}{z-1}$$

The domain is $1 < |z-2| < 2$

radius.

↓

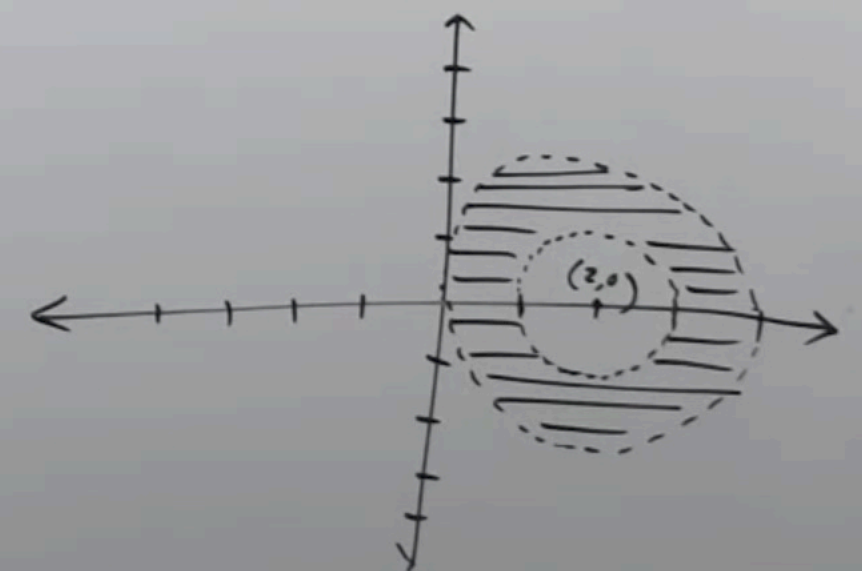
$$1 = |z-2|$$

centre

↓

$$|z-2| = 2 \quad \leftarrow \text{radius}$$

↑
centre



Ex: Expand $f(z) = \frac{1}{z(z-1)}$ as Laurent's Series in the domain $1 < |z-2| < 2$.

→

$$f(z) = \frac{-1}{2 \left[\left(\frac{z-2}{2} \right) + 1 \right]}$$

$$+ \frac{1}{(z-2) \left[1 + \frac{1}{z-2} \right]}$$

$$= \frac{-1}{2 \left[1 + \left(\frac{z-2}{2} \right) \right]} + \frac{1}{(z-2) \left[1 + \frac{1}{z-2} \right]}$$

$$= \frac{-1}{2} \times \frac{1}{\left(1 + \left(\frac{z-2}{2} \right) \right)} + \frac{1}{(z-2)} \times \frac{1}{\left[1 + \frac{1}{z-2} \right]}$$

$$= \frac{-1}{2} \left[1 - \left(\frac{z-2}{2} \right) + \left(\frac{z-2}{2} \right)^2 - \left(\frac{z-2}{2} \right)^3 + \dots \right]$$

$$+ \frac{1}{(z-2)} \left[1 - \frac{1}{z-2} + \frac{1}{(z-2)^2} - \frac{1}{(z-2)^3} + \dots \right]$$

$$\therefore f(z) = \frac{-1}{z} + \frac{1}{z-1}$$

The domain is $1 < |z-2| < 2$

ie. $1 < |z-2|$ & $|z-2| < 2$

$$\therefore \left| \frac{1}{z-2} \right| < 1 \quad \& \quad \left| \frac{z-2}{2} \right| < 1$$

We have,

$$f(z) = \frac{-1}{z} + \frac{1}{z-1}$$

$$= \frac{-1}{(z-2)+2} + \frac{1}{(z-2)+1}$$

$$\frac{1}{1-2} = 1 + 2 + 2^2 + 2^3 + \dots, |2| < 1$$

$$\frac{1}{1+2} = 1 - 2 + 2^2 - 2^3 + \dots, |2| < 1$$

